

TH77 Prime: Wiki Documentation

Lottery Management Platform
Designed & Developed by PSAI Studio
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1. Introduction

1.1 System Overview

TH77 Prime is a comprehensive Lottery Management Platform designed to streamline the operations of lottery organizations. It provides a secure and efficient environment for managing lottery rounds, betting, risk, and financials. The platform is built on a robust architecture that supports high-volume transactions and ensures data integrity and security.

- 4 Admin, Master, Agent, Member
- Thai Lottery, Lao Lottery, Hanoi Lottery, Stock Lottery
- Quick Key, Manual Selection
- Restricted Numbers, Risk Sharing
- Self-Service Betting Links
- Prize Calculation, Payouts, Settlements
- Bet History, Financial Reports, Agent Performance

1.2 Core Features (Key Features)

Feature Category	Feature Description
Multi-Tenant Architecture	Supports multiple tenants (Hierarchy: Admin → Master → Agent → Member)
Credit Management	Real-time credit limit management (e.g., 100,000 THB limit), credit freeze/thaw, credit transfer
Real-Time Processing	Real-time bet processing, real-time odds updates, real-time payout calculation
Link-Based Betting	Supports link-based betting (e.g., betting links for specific events)
Risk Control	Real-time risk monitoring, automated risk control rules, manual risk control
Automated Alerts	Automated alerts for system status, transaction status, etc. (e.g., transaction status=0)
OCR Integration	OCR integration for payment proof verification (Payment Proof)
High Availability	24/7 availability, auto-scaling

1.3 Technology Stack

Layer	Technology
Frontend	React (TypeScript), Next.js, Flutter (Mobile)
Backend	Node.js (Express/NestJS), Go (High-Throughput)
Database	PostgreSQL (Primary), Redis (Cache), MongoDB (Analytics)
Message Queue	RabbitMQ / Kafka
Search	Elasticsearch
Storage	MinIO (Object Storage)
API Gateway	Kong / Traefik
Monitoring	Prometheus, Grafana, Loki
CI/CD	GitHub Actions, Docker, Kubernetes
Security	JWT, RBAC, Cloudflare WAF, TLS

NO GGHG
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NO GGHG

Admin	
Master ()	
Agent ()	
Member ()	
3D	3 (: 123)
2D	2 (: 23)
Round	(: 123456)
Credit	
Credit Limit	(100,000 THB)
Restricted Number	()
Payout Rate	(: 1:800 3D)
Settlement	
Commission	/
Betting Link	
Audit Log	

[illegible][illegible]

Hardware Requirements

Component	Minimum	Recommended
CPU	4 Cores	8+ Cores
RAM	8 GB	16+ GB
Storage	50 GB SSD	100+ GB SSD

Component	Minimum	Recommended
Network	100 Mbps	1 Gbps

Software Requirements

Component	Version
Docker	20.10+
Docker Compose	2.0+
Kubernetes	1.25+
Node.js	18+
PostgreSQL	15+
Redis	7+
RabbitMQ	3.12+

3.2 📦📦📦📦📦📦📦📦📦📦 (Installation)

Local Development (Docker Compose)

1. Clone Repository

```
git clone https://github.com/your-repo/th77-prime.git
cd th77-prime
```

1. Setup Environment Variables

```
# Backend
cp backend/.env.example backend/.env
# Edit backend/.env with your settings

# Frontend
cp frontend/.env.example frontend/.env
# Edit frontend/.env with your settings
```

1. Start Services with Docker Compose

```
docker-compose up -d
```

1. Access the Application

- **Frontend:** <http://localhost:3001>
- **Backend API:** <http://localhost:3000/api/v1>
- **PGAdmin:** <http://localhost:5050>
- **RabbitMQ Management:** <http://localhost:15672>

3.3 📁📁📁📁📁📁📁📁📁📁 Environment Variables

Backend (.env)

```
# Server Configuration
NODE_ENV=development
PORT=3000

# Database Configuration
DB_HOST=postgres
DB_PORT=5432
DB_USER=postgres
DB_PASSWORD=postgres
DB_NAME=th77_prime

# JWT Configuration
JWT_SECRET=your_very_secure_jwt_secret_here
JWT_EXPIRES_IN=1d

# Redis Configuration
REDIS_HOST=redis
REDIS_PORT=6379

# RabbitMQ Configuration
RABBITMQ_HOST=rabbitmq
RABBITMQ_PORT=5672
RABBITMQ_USER=admin
RABBITMQ_PASSWORD=password

# External Services
LINE_CHANNEL_ACCESS_TOKEN=your_line_token
SMS_API_KEY=your_sms_key

# Logging
LOG_LEVEL=info
```

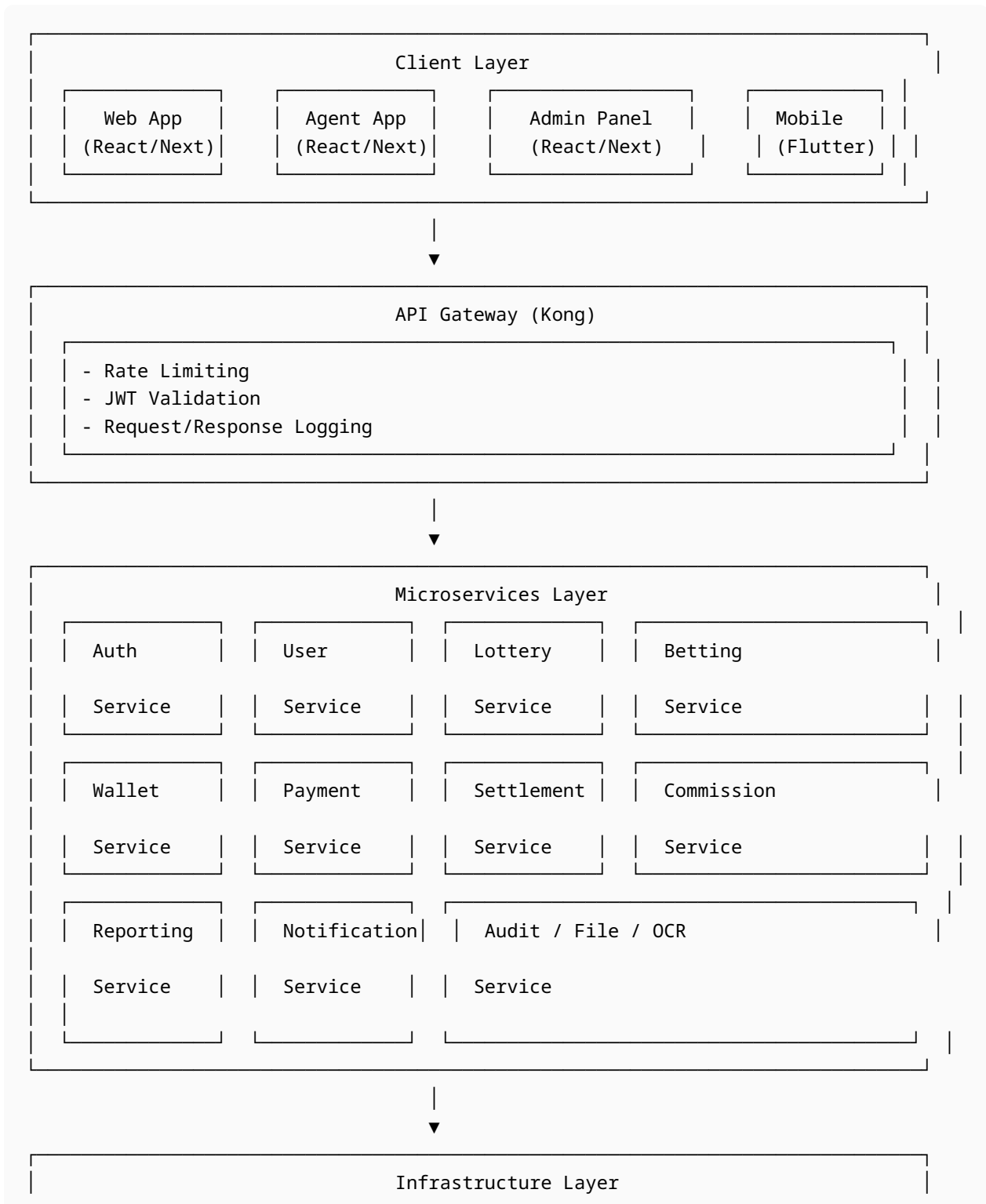
Frontend (.env)

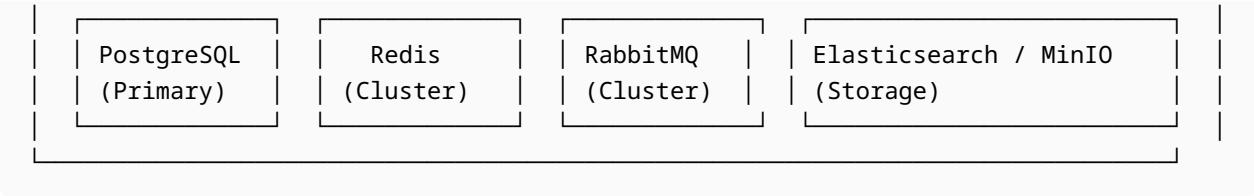
```
# API Configuration
NEXT_PUBLIC_API_BASE_URL=http://localhost:3000/api/v1
```

```
# App Configuration
NEXT_PUBLIC_APP_NAME=TH77 Prime
NEXT_PUBLIC_ENV=development
```

4. (System Architecture)

4.1 High-Level Architecture



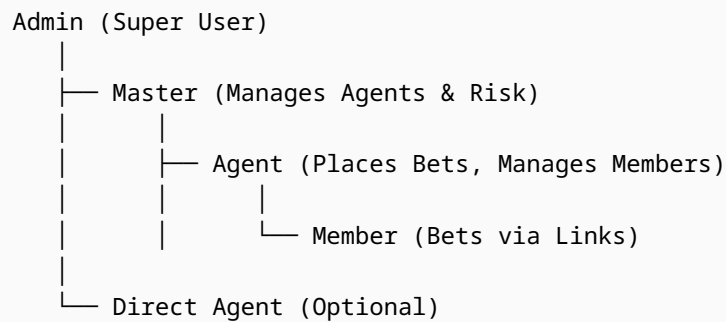


4.2 Service Breakdown

Service	Responsibility	Database	Dependencies
Auth Service	User authentication, JWT management	PostgreSQL	Redis (session)
User Service	User CRUD, role/permission management	PostgreSQL	Auth Service
Lottery Service	Lottery types, rounds, schedules	PostgreSQL	-
Betting Service	Bet placement, validation, duplication check	PostgreSQL	Lottery Service, Redis
Wallet Service	Credit management, deposits, withdrawals	PostgreSQL	User Service
Payment Service	Payment gateway integration, callbacks	PostgreSQL	Wallet Service
Settlement Service	Prize calculation, payout processing	PostgreSQL	Betting Service, Wallet Service
Commission Service	Agent hierarchy, commission calculation	PostgreSQL	User Service, Settlement Service
Reporting Service	Financial reports, bet history	PostgreSQL + MongoDB	All Services
Notification Service	SMS, Email, LINE notifications	-	RabbitMQ
Audit Service	Logging all financial operations	Elasticsearch	All Services
File Service	Document storage (payment proofs, OCR)	MinIO	-

[illegible]

5.1 Role Hierarchy



5.2 Permission Matrix

Action	Admin	Master	Agent	Member
User Management	 Create/Edit/Delete All	 Create/Edit Agents		
Credit Adjustment	 All Users	 Agents & Below	 Members Only	
Discount Management	 All Users	 Agents & Below	 Members Only	
Lottery/Round Management	 Full Access			
Bet Placement	 All	 All	 Own Bets	 Via Links
Bet Validation	 All	 All	 Own Bets	
Risk Management	 Full Access	 Set Restricted Numbers		
Link Generation	 All	 All	 Own Links	
Prize Calculation	 All	 All		
Payout Confirmation	 All	 All	 Own Payouts	
Financial Reports	 All	 All	 Own Reports	 Own Bets
Audit Logs	 All	 All		

5.3 User Workflow

Create User (Admin/Master)

1. Admin/Master
2. User Management

3. Add User
4. :
- Username

• Password

• Role (Agent/Member)

• Credit Limit (100,000 THB)

• Discount Percentage
1. Save

Adjust Credit (Admin/Master)

1. User
2. Adjust Credit
3. :
- • (/)

•
1. Confirm
2. Credit Transaction Audit Log

Block/Unblock User (Auto)

- = 0 →
- →

6. (Lottery & Round Management)

6.1 Lottery Types

Field	Description	Example
Name		Thai Lottery
Description		
Image URL		[URL]
Is Active		

Supported Lottery Types:

- Thai Lottery ()
- Lao Lottery ()

Manual Mode ()

- Select:     (3D/2D)
- Enter:  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  ,  , ,

7.4 Bet Types & Payout Rates

Bet Type	Description	Payout Rate	Example (100 THB Bet)
3D (Exact)	🎰 3 🎰🎰🎰	1:800	80,000 THB
2D (Exact)	🎰 2 🎰🎰	1:80	8,000 THB
3D (Any Order)	🎰 3 🎰🎰🎰	1:150	15,000 THB
2D (Any Order)	🎰 2 🎰🎰	1:40	4,000 THB

Note: Restricted Number (80% → 800 * 0.80 = 640 THB)

8. (Risk Management)

8.1 Restricted Numbers

Type	Description	Action
Blocked	██████████	🛑 ████████
Reduced Payout	██████ ████████████████	⚠️ ████████████████

Example:

- 824 → **Blocked** ()
- 123 → **Reduced Payout 80%** () (80%)

8.2 Risk Management Workflow

1. Admin/Master Sets Restricted Numbers

- [illegible]

1. Copy from Previous Round

- | |
|-----|
| 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 | 110 | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 | 121 | 122 | 123 | 124 | 125 | 126 | 127 | 128 | 129 | 130 | 131 | 132 | 133 | 134 | 135 | 136 | 137 | 138 | 139 | 140 | 141 | 142 | 143 | 144 | 145 | 146 | 147 | 148 | 149 | 150 | 151 | 152 | 153 | 154 | 155 | 156 | 157 | 158 | 159 | 160 | 161 | 162 | 163 | 164 | 165 | 166 | 167 | 168 | 169 | 170 | 171 | 172 | 173 | 174 | 175 | 176 | 177 | 178 | 179 | 180 | 181 | 182 | 183 | 184 | 185 | 186 | 187 | 188 | 189 | 190 | 191 | 192 | 193 | 194 | 195 | 196 | 197 | 198 | 199 | 200 | 201 | 202 | 203 | 204 | 205 | 206 | 207 | 208 | 209 | 210 | 211 | 212 | 213 | 214 | 215 | 216 | 217 | 218 | 219 | 220 | 221 | 222 | 223 | 224 | 225 | 226 | 227 | 228 | 229 | 230 | 231 | 232 | 233 | 234 | 235 | 236 | 237 | 238 | 239 | 240 | 241 | 242 | 243 | 244 | 245 | 246 | 247 | 248 | 249 | 250 | 251 | 252 | 253 | 254 | 255 | 256 | 257 | 258 | 259 | 260 | 261 | 262 | 263 | 264 | 265 | 266 | 267 | 268 | 269 | 270 | 271 | 272 | 273 | 274 | 275 | 276 | 277 | 278 | 279 | 280 | 281 | 282 | 283 | 284 | 285 | 286 | 287 | 288 | 289 | 290 | 291 | 292 | 293 | 294 | 295 | 296 | 297 | 298 | 299 | 300 | 301 | 302 | 303 | 304 | 305 | 306 | 307 | 308 | 309 | 310 | 311 | 312 | 313 | 314 | 315 | 316 | 317 | 318 | 319 | 320 | 321 | 322 | 323 | 324 | 325 | 326 | 327 | 328 | 329 | 330 | 331 | 332 | 333 | 334 | 335 | 336 | 337 | 338 | 339 | 340 | 341 | 342 | 343 | 344 | 345 | 346 | 347 | 348 | 349 | 350 | 351 | 352 | 353 | 354 | 355 | 356 | 357 | 358 | 359 | 360 | 361 | 362 | 363 | 364 | 365 | 366 | 367 | 368 | 369 | 370 | 371 | 372 | 373 | 374 | 375 | 376 | 377 | 378 | 379 | 380 | 381 | 382 | 383 | 384 | 385 | 386 | 387 | 388 | 389 | 390 | 391 | 392 | 393 | 394 | 395 | 396 | 397 | 398 | 399 | 400 | 401 | 402 | 403 | 404 | 405 | 406 | 407 | 408 | 409 | 410 | 411 | 412 | 413 | 414 | 415 | 416 | 417 | 418 | 419 | 420 | 421 | 422 | 423 | 424 | 425 | 426 | 427 | 428 | 429 | 430 | 431 | 432 | 433 | 434 | 435 | 436 | 437 | 438 | 439 | 440 | 441 | 442 | 443 | 444 | 445 | 446 | 447 | 448 | 449 | 450 | 451 | 452 | 453 | 454 | 455 | 456 | 457 | 458 | 459 | 460 | 461 | 462 | 463 | 464 | 465 | 466 | 467 | 468 | 469 | 470 | 471 | 472 | 473 | 474 | 475 | 476 | 477 | 478 | 479 | 480 | 481 | 482 | 483 | 484 | 485 | 486 | 487 | 488 | 489 | 490 | 491 | 492 | 493 | 494 | 495 | 496 | 497 | 498 | 499 | 500 | 501 | 502 | 503 | 504 | 505 | 506 | 507 | 508 | 509 | 510 | 511 | 512 | 513 | 514 | 515 | 516 | 517 | 518 | 519 | 520 | 521 | 522 | 523 | 524 | 525 | 526 | 527 | 528 | 529 | 530 | 531 | 532 | 533 | 534 | 535 | 536 | 537 | 538 | 539 | 540 | 541 | 542 | 543 | 544 | 545 | 546 | 547 | 548 | 549 | 550 | 551 | 552 | 553 | 554 |
|-----|

1. Monitor Risk in Real-Time

- HANNO 07
 HANNO 08
 HANNO 09
 HANNO 0A
 HANNO 0B
 HANNO 0C
 HANNO 0D
 HANNO 0E
 HANNO 0F
 HANNO 0G
 HANNO 0H
 HANNO 0I
 HANNO 0J
 HANNO 0K
 HANNO 0L
 HANNO 0M
 HANNO 0N
 HANNO 0O
 HANNO 0P
 HANNO 0Q
 HANNO 0R
 HANNO 0S
 HANNO 0T
 HANNO 0U
 HANNO 0V
 HANNO 0W
 HANNO 0X
 HANNO 0Y
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 HANNO 0~

-  (Risk Dashboard)

1. Send to Master Agent

- 
- 

1. Confirm from Master Agent

- 
-  Settlement

8.3 Risk Dashboard

Features:

- **Real-Time Monitoring** of bets per number
- **Color-Coded Risk Levels**
 - **Green:** Safe ( threshold)
 - **Yellow:** Caution ( threshold)
 - **Red:** High Risk ( threshold)
- **Automated Alerts** for high-risk numbers
- **One-Click Actions:**
 - Send to Master
 - Adjust Payout Rate
 - Block Number

9. (Link-Based Betting)

9.1 How It Works

1. Agent/Master Creates Link

- :
 - Customer Name
 - Expiry Time (15 , 1 , 1 )
 - Max Uses (/ / )
-  **Unique Token** (: abc123def456)

1. Share Link with Customer

- URL: <https://th77prime.com/bet/abc123def456>
-  LINE, SMS, Email

1. Customer Clicks Link

-  ()

- **Quick Key/Manual:**
 - **Quick Key:** Press the Quick Key to place a bet.
 - **Manual:** Press the Manual key to place a bet.
 - **Quick Key/Manual:** Press the Quick Key/Manual key to place a bet.

1. Customer Places Bet

- **Quick Key/Manual:** (Quick Key/Manual)
- **Quick Key/Manual:**
- **Quick Key/Manual:** = pending (Quick Key/Manual)
- **Quick Key/Manual:**

1. Agent Approves/Rejects Bet

- **Approve:** Quick Key/Manual = confirmed
- **Reject:** Quick Key/Manual = cancelled, Quick Key/Manual

9.2 Link Security

Feature	Description
Token-Based Access	Quick Key/Manual Token Quick Key/Manual
Expiry Time	Quick Key/Manual
Max Uses	Quick Key/Manual
IP Tracking	Quick Key/Manual IP Quick Key/Manual
Rate Limiting	Quick Key/Manual

9.3 Link Management

Action	Description
Create Link	Quick Key/Manual
List Links	Quick Key/Manual (Active/Expired)
Deactivate Link	Quick Key/Manual
View Bet History	Quick Key/Manual
Approve/Reject Bets	Quick Key/Manual/ Quick Key/Manual

[illegible]

10.1 Prize Calculation

Workflow:

1. Admin Sets Results

- (3D, 2D)

1. Run Calculation

- [illegible]

1. Review Winners

- [illegible]

10.2 Settlement Process

[illegible]

10.3 Financial Transactions

Type	Description	Effect on Credit
Bet	🎰🎰🎰🎰🎰	- (🎰🎰🎰🎰🎰🎰🎰🎰)
Deposit	💰💰💰💰💰💰💰💰💰	+
Withdrawal	💰💰💰💰💰💰💰💰	-

- **เครดิต:** 100,000 THB
- **ยอดคงเหลือ:** 94,960 (เครดิต: 94,960 - ค่าใช้จ่าย)
- **สถานะ:** 100,000 THB

12. API Documentation

12.1 Base URL & Authentication

- **Base URL:** `https://api.th77prime.com/v1`
- **Authentication:** Bearer <JWT_TOKEN> (Header: Authorization)
- **Content-Type:** `application/json`

12.2 Authentication API

Method	Endpoint	Description
POST	<code>/auth/login</code>	Login with username/password
POST	<code>/auth/refresh</code>	Refresh JWT token
POST	<code>/auth/logout</code>	Invalidate JWT token

Example: Login

```
curl -X POST https://api.th77prime.com/v1/auth/login \
-H "Content-Type: application/json" \
-d '{"username": "admin", "password": "secure123"}'
```

Response:

```
{
  "success": true,
  "data": {
    "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "refresh_token": "dGhpcyBpcyBhIHJlZnJlc2ggdG9rZW4...",
    "expires_in": 3600,
    "user": {
      "id": 1,
      "username": "admin",
      "role": "admin",
      "credit": 1000000.00
    }
  }
}
```

12.3 User Management API

Method	Endpoint	Description
GET	/users	List all users (paginated)
GET	/users/{id}	Get user details
POST	/users	Create a new user
PUT	/users/{id}	Update user details
DELETE	/users/{id}	Deactivate a user
PUT	/users/{id}/credit	Adjust user credit

Example: Adjust Credit

```
curl -X PUT https://api.th77prime.com/v1/users/3/credit \
-H "Authorization: Bearer <JWT_TOKEN>" \
-H "Content-Type: application/json" \
-d '{
  "amount": 50000.00,
  "action": "add",
  "reason": "Initial credit allocation"
}'
```

12.4 Lottery & Round API

Method	Endpoint	Description
GET	/lotteries	List all lottery types
POST	/lotteries	Create a new lottery type
GET	/rounds	List all rounds (filterable)
POST	/rounds	Create a new round
POST	/rounds/{id}/close	Close a round
POST	/rounds/{id}/results	Set round results
POST	/rounds/generate	Auto-generate upcoming rounds

Example: Set Round Results

```
curl -X POST https://api.th77prime.com/v1/rounds/101/results \
-H "Authorization: Bearer <JWT_TOKEN>" \
-H "Content-Type: application/json" \
-d '{
  "result_3d": "123",
  "result_2d": "23"
}'
```

12.5 Betting API

Method	Endpoint	Description
GET	/bets	List bets (filterable)
POST	/bets	Place a new bet
PUT	/bets/{id}	Update bet amount
DELETE	/bets/{id}	Cancel a bet
GET	/bets/{id}/validate	Validate bet before submission

Example: Place Bet

```
curl -X POST https://api.th77prime.com/v1/bets \
-H "Authorization: Bearer <JWT_TOKEN>" \
-H "Content-Type: application/json" \
-d '{
  "round_id": 101,
  "bets": [
    {
      "lottery_number": "824",
      "bet_type": "3d",
      "amount": 5040.00,
      "quantity": 1
    }
  ]
}'
```

12.6 Risk Management API

Method	Endpoint	Description
GET	/restricted-numbers	List restricted numbers
POST	/restricted-numbers	Add a restricted number

Method	Endpoint	Description
PUT	/restricted-numbers/{id}	Update restricted number
DELETE	/restricted-numbers/{id}	Remove restricted number
POST	/restricted-numbers/copy	Copy from previous round
GET	/risk/overview	Get risk overview

12.7 Link-Based Betting API

Method	Endpoint	Description
GET	/links	List all betting links
POST	/links	Create a new betting link
GET	/links/{token}	Get link details
POST	/links/{token}/bets	Place a bet via link

Example: Create Betting Link

```
curl -X POST https://api.th77prime.com/v1/links \
-H "Authorization: Bearer <JWT_TOKEN>" \
-H "Content-Type: application/json" \
-d '{
  "customer_name": "John Doe",
  "expiry_time": "2026-07-08T23:59:59Z",
  "max_uses": 5,
  "round_id": 101
}'
```

12.8 Financial API

Method	Endpoint	Description
GET	/transactions	List transactions
POST	/payouts/calculate	Calculate prizes for a round
POST	/payouts/confirm	Confirm prize payout
GET	/settlements	List settlements
POST	/settlements/{id}/confirm	Confirm settlement

12.9 Reporting API

Method	Endpoint	Description
GET	/reports/bets	Detailed bet reports
GET	/reports/financial	Financial summary
GET	/reports/agent	Agent performance reports
GET	/dashboard/overview	Overview dashboard

13. Database Schema

13.1 Core Tables

Users & Roles

```
CREATE TABLE roles (  
  id SERIAL PRIMARY KEY,  
  name VARCHAR(20) UNIQUE NOT NULL,  
  description TEXT  
);  
  
CREATE TABLE users (  
  id SERIAL PRIMARY KEY,  
  username VARCHAR(50) UNIQUE NOT NULL,  
  password_hash VARCHAR(255) NOT NULL,  
  email VARCHAR(100),  
  phone VARCHAR(20),  
  role_id INT REFERENCES roles(id),  
  parent_id INT REFERENCES users(id),  
  credit DECIMAL(15,2) DEFAULT 0.00,  
  max_credit_limit DECIMAL(15,2) DEFAULT 100000.00,  
  discount_percentage DECIMAL(5,2) DEFAULT 0.00,  
  is_active BOOLEAN DEFAULT true,  
  is_betting_blocked BOOLEAN DEFAULT false,  
  created_at TIMESTAMP DEFAULT NOW(),  
  updated_at TIMESTAMP DEFAULT NOW()  
);
```

Lottery & Rounds

```
CREATE TABLE lottery_types (  
  id SERIAL PRIMARY KEY,  
  name VARCHAR(50) NOT NULL,  
  description TEXT,
```

```

        image_url VARCHAR(255),
        is_active BOOLEAN DEFAULT true,
        created_at TIMESTAMP DEFAULT NOW()
    );

CREATE TABLE rounds (
    id SERIAL PRIMARY KEY,
    lottery_type_id INT REFERENCES lottery_types(id),
    round_number VARCHAR(20) NOT NULL,
    draw_date TIMESTAMP NOT NULL,
    close_time TIMESTAMP NOT NULL,
    open_time TIMESTAMP NOT NULL,
    result_3d VARCHAR(3),
    result_2d VARCHAR(2),
    is_closed BOOLEAN DEFAULT false,
    created_at TIMESTAMP DEFAULT NOW()
);

```

Betting System

```

CREATE TABLE bets (
    id SERIAL PRIMARY KEY,
    user_id INT REFERENCES users(id),
    round_id INT REFERENCES rounds(id),
    lottery_number VARCHAR(10) NOT NULL,
    bet_type VARCHAR(10) NOT NULL,
    amount DECIMAL(10,2) NOT NULL,
    total_amount DECIMAL(10,2) NOT NULL,
    quantity INT DEFAULT 1,
    discount_applied DECIMAL(5,2) DEFAULT 0.00,
    is_restricted BOOLEAN DEFAULT false,
    restricted_rate DECIMAL(5,2),
    status VARCHAR(20) DEFAULT 'pending',
    is_from_link BOOLEAN DEFAULT false,
    link_id INT REFERENCES betting_links(id),
    created_at TIMESTAMP DEFAULT NOW(),
    updated_at TIMESTAMP DEFAULT NOW()
);

```

Risk Management

```

CREATE TABLE restricted_numbers (
    id SERIAL PRIMARY KEY,
    round_id INT REFERENCES rounds(id),
    number VARCHAR(10) NOT NULL,
    is_blocked BOOLEAN DEFAULT true,
    payout_rate DECIMAL(5,2),
    created_by INT REFERENCES users(id),
    created_at TIMESTAMP DEFAULT NOW()
);

```

Financials

```
CREATE TABLE transactions (  
  id SERIAL PRIMARY KEY,  
  user_id INT REFERENCES users(id),  
  type VARCHAR(20) NOT NULL,  
  amount DECIMAL(15,2) NOT NULL,  
  balance_before DECIMAL(15,2) NOT NULL,  
  balance_after DECIMAL(15,2) NOT NULL,  
  reference_id INT,  
  reference_type VARCHAR(20),  
  status VARCHAR(20) DEFAULT 'completed',  
  proof_url VARCHAR(255),  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

Link-Based Betting

```
CREATE TABLE betting_links (  
  id SERIAL PRIMARY KEY,  
  created_by INT REFERENCES users(id),  
  customer_name VARCHAR(100),  
  token VARCHAR(255) UNIQUE NOT NULL,  
  expiry_time TIMESTAMP NOT NULL,  
  max_uses INT,  
  current_uses INT DEFAULT 0,  
  round_id INT REFERENCES rounds(id),  
  is_active BOOLEAN DEFAULT true,  
  created_at TIMESTAMP DEFAULT NOW()  
);  
  
CREATE TABLE link_bets (  
  id SERIAL PRIMARY KEY,  
  link_id INT REFERENCES betting_links(id),  
  bet_id INT REFERENCES bets(id),  
  customer_ip VARCHAR(45),  
  status VARCHAR(20) DEFAULT 'pending',  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

Settlements & Commissions

```
CREATE TABLE settlements (  
  id SERIAL PRIMARY KEY,  
  round_id INT REFERENCES rounds(id),  
  user_id INT REFERENCES users(id),  
  total_bet_amount DECIMAL(15,2) NOT NULL,  
  total_prize_amount DECIMAL(15,2) NOT NULL,  
  net_profit DECIMAL(15,2) NOT NULL,  
  commission_amount DECIMAL(15,2) DEFAULT 0.00,  
  status VARCHAR(20) DEFAULT 'pending',
```

```

        paid_at TIMESTAMP,
        created_at TIMESTAMP DEFAULT NOW()
    );

CREATE TABLE commissions (
    id SERIAL PRIMARY KEY,
    settlement_id INT REFERENCES settlements(id),
    user_id INT REFERENCES users(id),
    amount DECIMAL(15,2) NOT NULL,
    percentage DECIMAL(5,2) NOT NULL,
    status VARCHAR(20) DEFAULT 'pending',
    paid_at TIMESTAMP,
    created_at TIMESTAMP DEFAULT NOW()
);

```

Audit & Alerts

```

CREATE TABLE audit_logs (
    id SERIAL PRIMARY KEY,
    user_id INT REFERENCES users(id),
    action VARCHAR(50) NOT NULL,
    entity_type VARCHAR(50) NOT NULL,
    entity_id INT NOT NULL,
    old_value JSONB,
    new_value JSONB,
    ip_address VARCHAR(45),
    created_at TIMESTAMP DEFAULT NOW()
);

CREATE TABLE credit_alerts (
    id SERIAL PRIMARY KEY,
    user_id INT REFERENCES users(id),
    alert_type VARCHAR(20) NOT NULL,
    message TEXT NOT NULL,
    is_resolved BOOLEAN DEFAULT false,
    resolved_by INT REFERENCES users(id),
    resolved_at TIMESTAMP,
    created_at TIMESTAMP DEFAULT NOW()
);

```

13.2 Indexes for Performance

```

-- Users
CREATE INDEX idx_users_role ON users(role_id);
CREATE INDEX idx_users_parent ON users(parent_id);
CREATE INDEX idx_users_credit ON users(credit) WHERE credit <= 10000;

-- Rounds
CREATE INDEX idx_rounds_lottery_type ON rounds(lottery_type_id);
CREATE INDEX idx_rounds_draw_date ON rounds(draw_date);
CREATE INDEX idx_rounds_status ON rounds(is_closed);

```



```
-- Bets
CREATE INDEX idx_bets_user ON bets(user_id);
CREATE INDEX idx_bets_round ON bets(round_id);
CREATE INDEX idx_bets_status ON bets(status);
CREATE INDEX idx_bets_number_type ON bets(lottery_number, bet_type);
CREATE INDEX idx_bets_link ON bets(is_from_link, link_id);

-- Transactions
CREATE INDEX idx_transactions_user ON transactions(user_id);
CREATE INDEX idx_transactions_type ON transactions(type);
CREATE INDEX idx_transactions_status ON transactions(status);

-- Restricted Numbers
CREATE INDEX idx_restricted_round_number ON restricted_numbers(round_id, number);

-- Betting Links
CREATE INDEX idx_betting_links_token ON betting_links(token);
CREATE INDEX idx_betting_links_expiry ON betting_links(expiry_time);
```

14. Deploy (Deployment)

14.1 Docker Deployment (Local/Development)

1. Build Images

```
# Backend
cd backend
docker build -t th77prime/backend:v1.0.0 .

# Frontend
cd frontend
docker build -t th77prime/frontend:v1.0.0 .
```

1. Run with Docker Compose

```
docker-compose up -d
```

1. Access Services

- Frontend: <http://localhost:3001>
- Backend API: <http://localhost:3000/api/v1>
- PGAdmin: <http://localhost:5050>
- RabbitMQ: <http://localhost:15672>

14.2 Kubernetes Deployment (Production)

1. Build and Push Images

```
# Backend
cd backend
docker build -t th77prime/backend:v1.0.0 .
docker push th77prime/backend:v1.0.0

# Frontend
cd frontend
docker build -t th77prime/frontend:v1.0.0 .
docker push th77prime/frontend:v1.0.0
```

1. Apply Kubernetes Manifests

```
kubectl apply -f k8s/
```

1. Verify Deployment

```
kubectl get pods -n th77-prime
kubectl get svc -n th77-prime
kubectl get ingress -n th77-prime
```

1. Access Application

- Frontend: <https://th77prime.com>
- Backend API: <https://api.th77prime.com/api/v1>

14.3 Kubernetes Files

File	Description
k8s/namespace.yaml	Namespace Configuration
k8s/configmaps.yaml	Non-Sensitive Configuration
k8s/secrets.yaml	Sensitive Configuration (DB, JWT, etc.)
k8s/backend-deployment.yaml	Backend Deployment & Service
k8s/frontend-deployment.yaml	Frontend Deployment & Service
k8s/postgres-deployment.yaml	PostgreSQL StatefulSet & Service
k8s/redis-deployment.yaml	Redis Deployment & Service
k8s/rabbitmq-deployment.yaml	RabbitMQ StatefulSet & Service
k8s/ingress.yaml	Nginx Ingress with SSL
k8s/pvc.yaml	Persistent Volume Claims

File	Description
k8s/hpa.yaml	Horizontal Pod Autoscaler
k8s/kong-deployment.yaml	API Gateway (Kong)
k8s/monitoring.yaml	Prometheus, Grafana, Loki

15. 🧪🧪🧪🧪🧪🧪 (Testing)

15.1 Unit Testing

Backend (Jest)

```
cd backend
npm test
```

Example Test (bettingService.test.ts):

```
import { BetService } from '../services/bettingService';
import { User, Bet, Round } from '../models';

describe('BetService', () => {
  beforeAll(async () => {
    await User.create({ username: 'test-agent', role: 'agent', credit: 100000 });
    await Round.create({ round_number: '123456', draw_date: new Date(), is_closed:
false });
  });

  it('should place a bet and deduct credit', async () => {
    const result = await BetService.placeBet(1, 1, [
      { lottery_number: '123', bet_type: '3d', amount: 100, quantity: 1 }
    ]);

    expect(result.bets.length).toBe(1);
    expect(result.new_balance).toBe(99900);
  });

  it('should reject bet with insufficient credit', async () => {
    await expect(
      BetService.placeBet(1, 1, [
        { lottery_number: '456', bet_type: '3d', amount: 200000, quantity: 1 }
      ])
    ).rejects.toThrow('INSUFFICIENT_CREDIT');
  });
});
```

15.2 Integration Testing

Postman Collection:

- **Auth API** (Login, Refresh Token)
 - **User API** (CRUD, Credit Adjustment)
 - **Betting API** (Place Bet, Validate Bet)
 - **Risk API** (Restricted Numbers)
 - **Link API** (Create Link, Place Bet via Link)
-

15.3 End-to-End Testing (Cypress)

Example Test (betPlacement.spec.js):

```
describe('Bet Placement', () => {
  beforeEach(() => {
    cy.login('agent001', 'password123');
  });

  it('should place a bet successfully', () => {
    cy.visit('/betting');
    cy.get('#lottery-number').type('123');
    cy.get('#amount').type('100');
    cy.get('#quantity').type('1');
    cy.get('#place-bet').click();

    cy.contains('Bet placed successfully!');
    cy.contains('Credit: 99,900 THB');
  });

  it('should block betting when credit is zero', () => {
    // Simulate zero credit
    cy.intercept('POST', '/api/v1/bets', {
      statusCode: 400,
      body: { error: { code: 'ZERO_CREDIT' } }
    });

    cy.visit('/betting');
    cy.get('#place-bet').click();
    cy.contains('');
  });
});
```

[illegible]

16.1 Common Issues & Solutions

Issue	Cause	Solution
Login fails	Incorrect credentials	Reset password or check username
Bets not appearing	Not confirmed by agent	Contact agent to approve
Payout not received	Prize calculation not run	Run calculation for the round
Link not working	Expired or max uses reached	Generate new link
Slow API responses	High server load	Check monitoring (Grafana)
Credit not deducting	Transaction failed	Check audit logs
Database connection error	Incorrect DB config	Verify <code>k8s/secrets.yaml</code>
Redis connection error	Incorrect Redis config	Verify <code>k8s/configmaps.yaml</code>

16.2 Error Codes

Code	HTTP Status	Description	Solution
INVALID_INPUT	400	Missing or invalid request data	Check request body
UNAUTHORIZED	401	Missing or invalid JWT	Re-login
FORBIDDEN	403	Insufficient permissions	Check user role
NOT_FOUND	404	Resource not found	Verify ID/endpoint
INVALID_BET	400	Bet violates rules	Check number/type
INSUFFICIENT_CREDIT	400	User has no credit	Top up credit
ZERO_CREDIT	400	Credit is zero	Top up credit
ROUND_CLOSED	400	Betting is closed	Wait for next round
RESTRICTED_NUMBER	400	Number is blocked	Choose another number
RATE_LIMITED	429	Too many requests	Wait and retry
SERVER_ERROR	500	Internal server error	Check logs

16.3 Logs & Debugging

View Pod Logs:

```
kubectl logs -f deployment/th77-prime-backend -n th77-prime
kubectl logs -f deployment/th77-prime-frontend -n th77-prime
```

View All Logs with Timestamp:

```
kubectl logs --since-time=2026-07-07T00:00:00Z -f deployment/th77-prime-backend -n th77-prime
```

Debug Inside Pod:

```
kubectl exec -it deployment/th77-prime-backend -n th77-prime -- /bin/sh
```

17. (FAQ)

17.1 General Questions

Q: TH77 Prime?

A: 

Q:

A:  **Flutter App**  (Android/iOS)  **Responsive Web App**  Browser

Q:

A: **Scalability**

Hardware **Kubernetes Configuration**

17.2 User Management

Q:

A:

1. Login  Admin/Master
2.  **User Management**
3.  **Add User**
4.           (Username, Password, Role, Credit Limit)
5.  **Save**

Q: **How to adjust credit?**

A:

1. **User**
 2. **Adjust Credit**
 3. **Account ID**, **Account Name** (**Account ID/Account Name**), **Account Type**
 4. **Confirm**
-

Q: **How to adjust credit?**

A: **Account ID**, **Account Name** (**Account ID/Account Name**), **Account Type**

17.3 Betting System

Q: **How to use betting system?**

A:

1. Login **Agent/Master**
 2. **Betting Page**
 3. **Quick Key** **Manual Mode**
 4. **Account ID**, **Account Name**, **Account Type**
 5. **Confirm**
-

Q: **How to use Quick Key?**

A: **Format:** [**Account ID**] [**Account Name**]x[**Account Type**]

Example: 824 5040x1, 123 100x2

Q: **How to use Quick Key?**

A: **Account ID**, **Account Name** (**Account ID/Account Name**): **Account ID** 123 100x1 **Account Name** 123 50x1 → **Account ID/Account Name** 123 150x1)

Q: **How to use Quick Key?**

A:

- **Blocked** → **Account ID/Account Name**
 - **Reduced Payout** → **Account ID/Account Name** (**Account ID/Account Name**: **Account ID/Account Name** 80%)
-

17.4 Link-Based Betting

Q: **How to use link-based betting?**

A:

1. Login **Agent/Master**
2. **Link Management**
3. **Create Link**
4. **Customer Name**, **Expiry Time**, **Max Uses**
5. **Generate Link**

6.

Q: **?**

A: **Max Uses** (: 5)

[illegible]

A:

17.5 Financial Processing

Q:

A:

1. Admin (3D, 2D)
2. Calculate Prizes
- 3.

[illegible]

A:

- **3D (Exact):** 1:800
- **2D (Exact):** 1:80
- **3D (Any Order):** 1:150
- **2D (Any Order):** 1:40

Note: **Restricted Number**

Q:

A:

1. **Settlement Page**
2. **Calculate Prizes**
3. **Confirm Payout**

17.6 Deployment

Q: Deploy?

A:

- ## 1. Build Docker Images


```
docker build -t th77prime/backend:v1.0.0 ./backend
docker build -t th77prime/frontend:v1.0.0 ./frontend
docker push th77prime/backend:v1.0.0
docker push th77prime/frontend:v1.0.0
```

1. Apply Kubernetes Manifests

```
kubectl apply -f k8s/
```

Q: ?

A:

1. **Update Code**
2. **Rebuild Docker Images**

```
docker build -t th77prime/backend:v1.0.1 ./backend
docker push th77prime/backend:v1.0.1
```

1. Update Kubernetes Deployment

```
kubectl set image deployment/th77-prime-backend backend=th77prime/backend:v1.0.1 -n th77-prime
```

Q:  Rollback?

A:

```
kubectl rollout undo deployment/th77-prime-backend -n th77-prime
```

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